Worksheet 6-3 B Integrals, Area, and Riemann Sums #1
AP Calculus Name_____

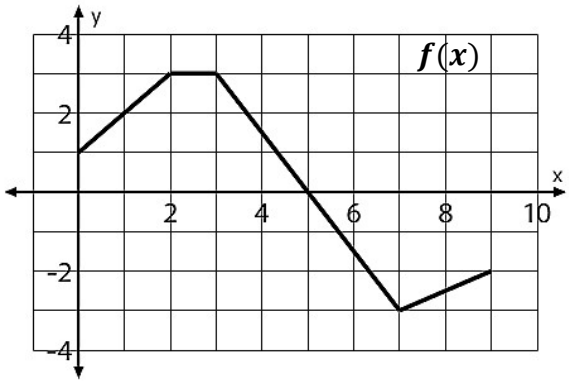
The graph of $f(x)$ is shown. Evaluate each integral by interpreting it in terms of areas.

1.) $\int_0^2 f(x)\,dx$

2.) $\int_0^5 f(x)\,dx$

3.) $\int_5^7 f(x)\,dx$

4.) $\int_0^9 f(x)\,dx$

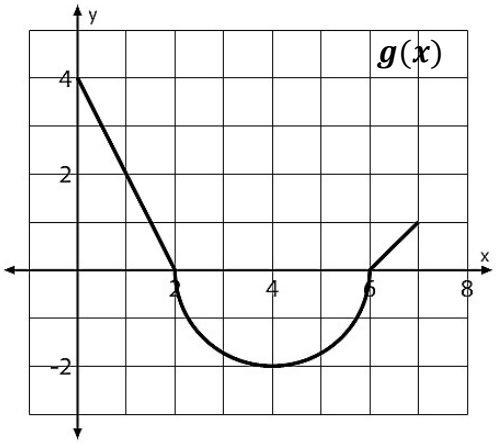


The graph of $g(x)$ at the right consists of two straight lines and a semi-circle. Use it to evaluate each integral.

5.) $\int_0^2 g(x)\,dx$

6.) $\int_2^6 g(x)\,dx$

7.) $\int_0^7 g(x)\,dx$



Evaluate the integral by graphing the function, then interpreting the integral in terms of areas.

8.) $\int_{-1}^2 (1-x)\,dx$

9.) $\int_0^9 \left(\frac{1}{3}x - 2\right)\,dx$

Odie goes out for a 90-minute run. As is his pattern, he starts out too fast and then slows down. The table at the right gives his speed in miles per hour after a given number of minutes. Pay close attention to the units!

Time (min)	0	15	30	45	60	75	90
Speed (mi/hr)	11	10	10	9	7	6	0

10.) Approximate the distance that Odie travels using a Midpoint Riemann sum with 3 equal subintervals. Be sure to include units in your answer.

11.) Approximate the distance that Odie travels using a Trapezoidal approximation with 6 trapezoids. Be sure to include units in your answer.

Match each definite integral with the appropriate limit of a Riemann Sum in the right column.

12. $\int_1^3 (4x^2+2)dx$

13. $\int_2^5 (x^3+1)dx$

14. $\int_2^4 (4x^2+2)dx$

15. $\int_2^5 (4x^2+2)dx$

16. $\int_3^1 (4x^2+2)dx$

17. $\int_5^7 (x^3+1)dx$

A. $\lim_{n\to\infty}\sum_{j=1}^{n}\left[4\left(2+\frac{3j}{n}\right)^2+2\right]\left(\frac{3}{n}\right)$

B. $\lim_{n\to\infty}\sum_{j=1}^{n}\left[\left(2+\frac{3j}{n}\right)^3+1\right]\left(\frac{3}{n}\right)$

C. $\lim_{n\to\infty}\sum_{j=1}^{n}\left[4\left(1+\frac{2j}{n}\right)^2+2\right]\left(\frac{2}{n}\right)$

D. $\lim_{n\to\infty}\sum_{j=1}^{n}\left[4\left(3-\frac{2j}{n}\right)^2+2\right]\left(\frac{-2}{n}\right)$

E. $\lim_{n\to\infty}\sum_{j=1}^{n}\left[\left(5+\frac{2j}{n}\right)^3+1\right]\left(\frac{2}{n}\right)$

F. $\lim_{n\to\infty}\sum_{j=1}^{n}\left[4\left(2+\frac{2j}{n}\right)^2+2\right]\left(\frac{2}{n}\right)$